

```

deepakk@deepakk-VMware-Virtual-Platform: ~
provide /usr/bin/jrunscript (jrunscript) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jshell to provide /usr/bin/jshell (jshell) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jstack to provide /usr/bin/jstack (jstack) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jstat to provide /usr/bin/jstat (jstat) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jstatd to provide /usr/bin/jstatd (jstatd) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/serialver to provide /usr/bin/serialver (serialver) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jhsdb to provide /usr/bin/jhsdb (jhsdb) in auto mode
Setting up openjdk-17-jdk:amd64 (17.0.19+10-1-26.04.2)...
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jconsole to provide /usr/bin/jconsole (jconsole) in auto mode
deepakk@deepakk-VMware-Virtual-Platform:~$ readlink -f $(which java)
/usr/lib/jvm/java-17-openjdk-amd64/bin/java
deepakk@deepakk-VMware-Virtual-Platform:~$ java -version
openjdk version "17.0.19" 2026-04-21
OpenJDK Runtime Environment (build 17.0.19+10-1-26.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 17.0.19+10-1-26.04.2-Ubuntu, mixed mode, sharing)
deepakk@deepakk-VMware-Virtual-Platform:~$

```

```

deepakk@deepakk-VMware-Virtual-Platform: ~ — su...
em/ssh.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/ssh.service' → '/usr/lib/systemd/system/ssh.service'.
deepakk@deepakk-VMware-Virtual-Platform:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-08-25 20:14:48 IST; 36s ago
     Invocation: ff3bf193d1b748b48bc82b7a6fa43643
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
    Process: 6655 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 6657 (sshd)
      Tasks: 1 (limit: 1856)
     Memory: 3M (peak: 3.8M)
        CPU: 131ms
     CGroup: /system.slice/ssh.service
             └─6657 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Aug 25 20:14:48 deepakk-VMware-Virtual-Platform systemd[1]: Starting ssh.service:
Aug 25 20:14:48 deepakk-VMware-Virtual-Platform sshd[6657]: Server listening on
Aug 25 20:14:48 deepakk-VMware-Virtual-Platform sshd[6657]: Server listening on
Aug 25 20:14:48 deepakk-VMware-Virtual-Platform systemd[1]: Started ssh.service:
lines 1-19/19 (END)

```

```

deepakk@deepakk-VMware-Virtual-Platform: ~
Aug 25 20:14:48 deepakk-VMware-Virtual-Platform sshd[6657]: Server listening on
Aug 25 20:14:48 deepakk-VMware-Virtual-Platform systemd[1]: Started ssh.service

deepakk@deepakk-VMware-Virtual-Platform:~$ ssh-keygen -t rsa -P ""
Generating public/private rsa key pair.
Enter file in which to save the key (/home/deepakk/.ssh/id_rsa):
Your identification has been saved in /home/deepakk/.ssh/id_rsa
Your public key has been saved in /home/deepakk/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:liFCi8/Cf4GDCymX+rGLwTriGiaQXMBVdbdl8HiEZYg deepakk@deepakk-VMware-Virtua
l-Platform
The key's randomart image is:
+---[RSA 3072]-----+
|..o+.... ..o=*|
|+.B o .E.OO|
|+*. * o . o o|
|O=+. o o o .|
|+.o . . S|
|o. o . .|
|+oo|
|*+ .|
|Bo.|
+-----[SHA256]-----+
deepakk@deepakk-VMware-Virtual-Platform:~$

```

```

deepakk@deepakk-VMware-Virtual-Platform: ~ -- ss...
deepakk@deepakk-VMware-Virtual-Platform:~$ chmod 600 ~/.ssh/authorized_keys
deepakk@deepakk-VMware-Virtual-Platform:~$ ssh localhost
The authenticity of host 'localhost (127.0.0.1)' can't be established.
ED25519 key fingerprint is: SHA256:XezWBij7y7r/3i+UiMo1wvFB6GouGUQFD9MCI/0RBAU
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-14-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

319 updates can be applied immediately.
217 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the

```

```
deepakk@deepakk-VMware-Virtual-Platform: ~ — ss...
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

deepakk@deepakk-VMware-Virtual-Platform:~$ exit
logout
Connection to localhost closed.
deepakk@deepakk-VMware-Virtual-Platform:~$ ssh localhost
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-14-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

319 updates can be applied immediately.
217 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Tue Aug 25 20:18:26 2026 from 127.0.0.1
deepakk@deepakk-VMware-Virtual-Platform:~$
```

```
p/common
deepakk@deepakk-VMware-Virtual-Platform: ~
Length: 581280703 (554M) [application/x-gzip]
Saving to: 'hadoop-3.5.0.tar.gz'

0 hadoop-3.5.0.tar.gz 17%[==>] 94.85M 152KB/s eta 47m 7s ^
0 hadoop-3.5.0.tar.gz 17%[==>] 95.01M 130KB/s eta 47m 7s ^
0 hadoop-3.5.0.tar.gz 17%[==>] 95.14M 112KB/s eta 47m 10s ^
0 hadoop-3.5.0.tar.gz 17%[==>] 95.20M 98.8KB/s eta 47m 10s ^
0 hadoop-3.5.0.tar.gz 17%[==>] 95.71M 96.1KB/s eta 47m 19s ^
0 hadoop-3.5.0.tar.gz 17%[==>] 95.79M 95.2KB/s eta 47m 20s ^
0 hadoop-3.5.0.tar.gz 17%[==>] 95.82M 92.3KB/s eta 47m 20s ^
0 hadoop-3.5.0.tar.gz 23%[===>] 129.00M 181KB/s eta 44m 41s ^
0 hadoop-3.5.0.tar.gz 23%[===>] 129.15M 187KB/s eta 44m 39s ^
0 hadoop-3.5.0.tar.gz 100%[=====] 554.35M 493KB/s in 32m 28s

2026-08-25 21:04:33 (291 KB/s) - 'hadoop-3.5.0.tar.gz' saved [581280703/581280703]

0 deepakk@deepakk-VMware-Virtual-Platform:~$ ls -lh hadoop-3.5.0.tar.gz
-rw-rw-r-- 1 deepakk deepakk 555M Apr  2 02:13 hadoop-3.5.0.tar.gz
0 deepakk@deepakk-VMware-Virtual-Platform:~$ tar -xzf hadoop-3.5.0.tar.gz
0 deepakk@deepakk-VMware-Virtual-Platform:~$ sudo mv hadoop-3.5.0 /usr/local/hadoop
2 p
2 [sudo: authenticate] Password:
2 deepakk@deepakk-VMware-Virtual-Platform:~$
```



```

deepakk@deepakk-VMware-Virtual-Platform: ~
hadoop-3.5.0.tar.gz 17%[==>] 94.85M 152KB/s eta 47m 7s ^
hadoop-3.5.0.tar.gz 17%[==>] 95.01M 130KB/s eta 47m 7s ^
hadoop-3.5.0.tar.gz 17%[==>] 95.14M 112KB/s eta 47m 10s^
hadoop-3.5.0.tar.gz 17%[==>] 95.20M 98.8KB/s eta 47m 10s^
hadoop-3.5.0.tar.gz 17%[==>] 95.71M 96.1KB/s eta 47m 19s^
hadoop-3.5.0.tar.gz 17%[==>] 95.79M 95.2KB/s eta 47m 20s^
hadoop-3.5.0.tar.gz 17%[==>] 95.82M 92.3KB/s eta 47m 20s^
hadoop-3.5.0.tar.gz 23%[====>] 129.00M 181KB/s eta 44m 41s^
hadoop-3.5.0.tar.gz 23%[====>] 129.15M 187KB/s eta 44m 39s^
hadoop-3.5.0.tar.gz 100%[=====>] 554.35M 493KB/s in 32m 28s

2026-08-25 21:04:33 (291 KB/s) - 'hadoop-3.5.0.tar.gz' saved [581280703/58128070
3]

deepakk@deepakk-VMware-Virtual-Platform:~$ ls -lh hadoop-3.5.0.tar.gz
-rw-rw-r-- 1 deepakk deepakk 555M Apr  2 02:13 hadoop-3.5.0.tar.gz
deepakk@deepakk-VMware-Virtual-Platform:~$ tar -xzf hadoop-3.5.0.tar.gz
deepakk@deepakk-VMware-Virtual-Platform:~$ sudo mv hadoop-3.5.0 /usr/local/hadoo
p
[sudo: authenticate] Password:
deepakk@deepakk-VMware-Virtual-Platform:~$ /usr/local/hadoop/bin/hadoop version
ERROR: JAVA_HOME is not set and could not be found.
deepakk@deepakk-VMware-Virtual-Platform:~$

```

```

deepakk@deepakk-VMware-Virtual-Platform: ~
deepakk@deepakk-VMware-Virtual-Platform:~$ /usr/local/hadoop/bin/hadoop version
ERROR: JAVA_HOME is not set and could not be found.
deepakk@deepakk-VMware-Virtual-Platform:~$ java -version
openjdk version "17.0.19" 2026-04-21
OpenJDK Runtime Environment (build 17.0.19+10-1-26.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 17.0.19+10-1-26.04.2-Ubuntu, mixed mode, sharing
)
deepakk@deepakk-VMware-Virtual-Platform:~$ readlink -f $(which java)
/usr/lib/jvm/java-17-openjdk-amd64/bin/java
deepakk@deepakk-VMware-Virtual-Platform:~$ export JAVA_HOME=/usr/lib/jvm/java-17
-openjdk-amd64
deepakk@deepakk-VMware-Virtual-Platform:~$ echo $JAVA_HOME
/usr/lib/jvm/java-17-openjdk-amd64
deepakk@deepakk-VMware-Virtual-Platform:~$ /usr/local/hadoop/bin/hadoop version
Hadoop 3.5.0
Source code repository https://github.com/apache/hadoop.git -r dbcc7cd797100e6b3
2cd84f85b53a5193a5f9af0
Compiled by cnauroth on 2026-03-24T16:56Z
Compiled on platform linux-x86_64
Compiled with protoc 3.25.5
From source with checksum 64582ed0a62a169c39a19c734a8f33b5
This command was run using /usr/local/hadoop/share/hadoop/common/hadoop-common-3
.5.0.jar
deepakk@deepakk-VMware-Virtual-Platform:~$

```

```
deepakk@deepakk-VMware-Virtual-Platform: ~  
deepakk@deepakk-VMware-Virtual-Platform:~$ sudo nano /usr/local/hadoop/etc/hadoop/hadoop-env.sh  
[sudo: authenticate] Password:  
deepakk@deepakk-VMware-Virtual-Platform:~$ nano ~/.bashrc  
deepakk@deepakk-VMware-Virtual-Platform:~$ source ~/.bashrc  
deepakk@deepakk-VMware-Virtual-Platform:~$ echo $JAVA_HOME  
/usr/lib/jvm/java-17-openjdk-amd64  
deepakk@deepakk-VMware-Virtual-Platform:~$ echo $HADOOP_HOME  
/usr/local/hadoop  
deepakk@deepakk-VMware-Virtual-Platform:~$ hadoop version  
Hadoop 3.5.0  
Source code repository https://github.com/apache/hadoop.git -r dbcc7cd797100e6b32cd84f85b53a5193a5f9af0  
Compiled by cnauroth on 2026-03-24T16:56Z  
Compiled on platform linux-x86_64  
Compiled with protoc 3.25.5  
From source with checksum 64582ed0a62a169c39a19c734a8f33b5  
This command was run using /usr/local/hadoop/share/hadoop/common/hadoop-common-3.5.0.jar  
deepakk@deepakk-VMware-Virtual-Platform:~$
```

```
deepakk@deepakk-VMware-Virtual-Platform:~$ sudo nano /usr/local/hadoop/etc/hadoop/core-site.xml  
deepakk@deepakk-VMware-Virtual-Platform:~$ sudo nano /usr/local/hadoop/etc/hadoop/hdfs-site.xml  
deepakk@deepakk-VMware-Virtual-Platform:~$
```

```
deepakk@deepakk-VMware-Virtual-Platform: ~ — su...  
GNU nano 8.7.1 /usr/local/hadoop/etc/hadoop/core-site.xml  
<?xml version="1.0"?>  
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>  
  
<configuration>  
  
  <property>  
    <name>fs.defaultFS</name>  
    <value>hdfs://localhost:9000</value>  
  </property>  
  
</configuratio>  
  
[ Read 11 lines ]  
^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute  ^C Location  
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify  ^_ Go To Line
```

```
deepakk@deepakk-VMware-Virtual-Platform: ~  
deepakk@deepakk-VMware-Virtual-Platform:~$ sudo mkdir -p /usr/local/hadoop/data/  
namenode  
[sudo: authenticate] Password:  
deepakk@deepakk-VMware-Virtual-Platform:~$ sudo mkdir -p /usr/local/hadoop/data/  
datanode  
deepakk@deepakk-VMware-Virtual-Platform:~$ sudo chown -R $USER:$USER /usr/local/  
hadoop/data  
deepakk@deepakk-VMware-Virtual-Platform:~$ ls -l /usr/local/hadoop/data  
total 8  
drwxr-xr-x 2 deepakk deepakk 4096 Aug 25 22:07 datanode  
drwxr-xr-x 2 deepakk deepakk 4096 Aug 25 22:07 namenode  
deepakk@deepakk-VMware-Virtual-Platform:~$  
  
deepakk@deepakk-VMware-Virtual-Platform:~$ sudo nano /usr/local/hadoop/etc/hadoo  
p/mapred-site.xml  
deepakk@deepakk-VMware-Virtual-Platform:~$ sudo nano /usr/local/hadoop/etc/hadoo  
p/mapred-site.xml
```

```
GNU nano 8.7.1 /usr/local/hadoop/etc/hadoo  
p/mapred-site.xml  
<?xml version="1.0"?>  
<?xml-stylesheet type="text/xsl" href="configur  
  
<configuration>  
  <property>  
    <name>mapreduce.framework.name</name>  
    <value>yarn</value>  
  </property>  
  
</configuration>
```

```
deepakk@deepakk-VMware-Virtual-Platform: ~  
deepakk@deepakk-VMware-Virtual-Platform:~$ sudo nano /usr/local/hadoop/etc/hadoop/yarn-site.xml  
[sudo: authenticate] Password:  
deepakk@deepakk-VMware-Virtual-Platform:~$ sudo nano /usr/local/hadoop/etc/hadoop/yarn-site.xml  
deepakk@deepakk-VMware-Virtual-Platform:~$
```

```
deepakk@deepakk-VMware-Virtual-Platform: ~ — su...  
GNU nano 8.7.1 /usr/local/hadoop/etc/hadoop/yarn-site.xml  
<?xml version="1.0"?>  
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>  
  
<configuration>  
  <property>  
    <name>yarn.nodemanager.aux-services</name>  
    <value>mapreduce_shuffle</value>  
  </property>  
</configuration>  
  
[ Read 11 lines ]  
^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location  
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line  
  
neNode  
deepakk@deepakk-VMware-Virtual-Platform:~$ hdfs namenode -format  
WARNING: /usr/local/hadoop/logs does not exist. Creating.
```



```
deepakk@deepakk-VMware-Virtual-Platform:~$ hdfs namenode -format
WARNING: /usr/local/hadoop/logs does not exist. Creating.
2026-08-25 22:12:01,527 INFO namenode.NameNode: STARTUP_MSG:
/*****
STARTUP_MSG: Starting NameNode
STARTUP_MSG: host = deepakk-VMware-Virtual-Platform/127.0.1.1
STARTUP_MSG: args = [-format]
STARTUP_MSG: version = 3.5.0
STARTUP_MSG: classpath = /usr/local/hadoop/etc/hadoop:/usr/local/hadoop/share/
hadoop/common/lib/jetty-server-9.4.58.v20250814.jar:/usr/local/hadoop/share/hado
op/common/lib/commons-daemon-1.0.13.jar:/usr/local/hadoop/share/hadoop/common/li
b/aopalliance-repackaged-2.6.1.jar:/usr/local/hadoop/share/hadoop/common/lib/jet
ty-xml-9.4.58.v20250814.jar:/usr/local/hadoop/share/hadoop/common/lib/commons-co
nfiguration2-2.10.1.jar:/usr/local/hadoop/share/hadoop/common/lib/jakarta.servle
t-api-4.0.4.jar:/usr/local/hadoop/share/hadoop/common/lib/azure-storage-7.0.1.ja
r:/usr/local/hadoop/share/hadoop/common/lib/commons-net-3.9.0.jar:/usr/local/had
oop/share/hadoop/common/lib/hk2-utils-2.6.1.jar:/usr/local/hadoop/share/hadoop/c
```

```
... 10 more
deepakk@deepakk-VMware-Virtual-Platform:~$ start-dfs.sh
Starting namenodes on [deepakk-VMware-Virtual-Platform]
deepakk-VMware-Virtual-Platform: Warning: Permanently added 'deepakk-vmware-virt
ual-platform' (ED25519) to the list of known hosts.
deepakk-VMware-Virtual-Platform: ERROR: JAVA_HOME is not set and could not be fo
und.
Starting datanodes
localhost: ERROR: JAVA_HOME is not set and could not be found.
2026-08-25 22:13:10,200 ERROR conf.Configuration: error parsing conf core-site.x
ml
com.ctc.wstx.exc.WstxParsingException: Unexpected close tag </configuratio>; exp
ected </configuration>.
  at [row,col,system-id]: [11,14,"file:/usr/local/hadoop/etc/hadoop/core-site.xml
"]
    at com.ctc.wstx.sr.StreamScanner.constructWfcException(StreamScanner.jav
a:634)
    at com.ctc.wstx.sr.StreamScanner.throwParseError(StreamScanner.java:504)
    at com.ctc.wstx.sr.StreamScanner.throwParseError(StreamScanner.java:488)
    at com.ctc.wstx.sr.BasicStreamReader.reportWrongEndElement(BasicStreamReade
r.java:3352)
```

```
deepakk@deepakk-VMware-Virtual-Platform:~$ jps
54483 Jps
deepakk@deepakk-VMware-Virtual-Platform:~$
```



```
[200~cat: command not found
deepakk@deepakk-VMware-Virtual-Platform:~$ sudo nano /usr/local/hadoop/etc
/hadoop/core-site.xml
deepakk@deepakk-VMware-Virtual-Platform:~$ hdfs getconf -confKey fs.defaultFS
hdfs://localhost:9000
deepakk@deepakk-VMware-Virtual-Platform:~$ hdfs getconf -confKey fs.defaultFS
hdfs://localhost:9000
deepakk@deepakk-VMware-Virtual-Platform:~$ sudo nano /usr/local/hadoop/etc
/hadoop/hdfs-site.xml
deepakk@deepakk-VMware-Virtual-Platform:~$ sudo mkdir -p /usr/local/hadoop
/data/namenode
deepakk@deepakk-VMware-Virtual-Platform:~$ sudo mkdir -p /usr/local/hadoop
/data/datanode
deepakk@deepakk-VMware-Virtual-Platform:~$ sudo chown -R $USER:$USER /usr/
local/hadoop/data
deepakk@deepakk-VMware-Virtual-Platform:~$ cat /usr/local/hadoop/etc/hadoo
p/hdfs-site.xml
<?xml version="1.0"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>

<configuration>

  <property>
    <name>dfs.replication</name>
    <value>1</value>
  </property>
```

```
/data/datanode
deepakk@deepakk-VMware-Virtual-Platform:~$ sudo chown -R $USER:$USER /usr/
local/hadoop/data
deepakk@deepakk-VMware-Virtual-Platform:~$ cat /usr/local/hadoop/etc/hadoo
p/hdfs-site.xml
<?xml version="1.0"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>

<configuration>

  <property>
    <name>dfs.replication</name>
    <value>1</value>
  </property>

  <property>
    <name>dfs.namenode.name.dir</name>
    <value>file:///usr/local/hadoop/data/namenode</value>
  </property>

  <property>
    <name>dfs.datanode.data.dir</name>
    <value>file:///usr/local/hadoop/data/datanode</value>
  </property>

</configuration>
deepakk@deepakk-VMware-Virtual-Platform:~$ sudo nano /usr/local/hadoop/etc
/hadoop/mapred-site.xml
deepakk@deepakk-VMware-Virtual-Platform:~$ sudo nano /usr/local/hadoop/etc
/hadoop/yarn-site.xml
deepakk@deepakk-VMware-Virtual-Platform:~$ cat /usr/local/hadoop/etc/hadoo
p/mapred-site.xml
<?xml version="1.0"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>

<configuration>

  <property>
    <name>mapreduce.framework.name</name>
    <value>yarn</value>
```

```

</property>

<property>
  <name>dfs.datanode.data.dir</name>
  <value>file:///usr/local/hadoop/data/datanode</value>
</property>

</configuration>
deepakk@deepakk-VMware-Virtual-Platform:~$ sudo nano /usr/local/hadoop/etc
/hadoop/mapred-site.xml
deepakk@deepakk-VMware-Virtual-Platform:~$ sudo nano /usr/local/hadoop/etc
/hadoop/yarn-site.xml
deepakk@deepakk-VMware-Virtual-Platform:~$ cat /usr/local/hadoop/etc/hadoo
p/mapred-site.xml
<?xml version="1.0"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>

<configuration>

  <property>
    <name>mapreduce.framework.name</name>
    <value>yarn</value>
  </property>

</configuration>
deepakk@deepakk-VMware-Virtual-Platform:~$ cat /usr/local/hadoop/etc/hadoo
p/mapred-site.xml
<?xml version="1.0"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>

<configuration>

  <property>
    <name>mapreduce.framework.name</name>
    <value>yarn</value>
  </property>

</configuration>
deepakk@deepakk-VMware-Virtual-Platform:~$ █
</configuration>
deepakk@deepakk-VMware-Virtual-Platform:~$ hdfs getconf -confKey fs.default
tFS
hdfs://localhost:9000
deepakk@deepakk-VMware-Virtual-Platform:~$ hdfs getconf -confKey dfs.repli
cation
1
deepakk@deepakk-VMware-Virtual-Platform:~$ hadoop version
Hadoop 3.5.0
Source code repository https://github.com/apache/hadoop.git -r dbcc7cd7971
00e6b32cd84f85b53a5193a5f9af0
Compiled by cnauroth on 2026-03-24T16:56Z
Compiled on platform linux-x86_64
Compiled with protoc 3.25.5
From source with checksum 64582ed0a62a169c39a19c734a8f33b5
This command was run using /usr/local/hadoop/share/hadoop/common/hadoop-co
mmon-3.5.0.jar
deepakk@deepakk-VMware-Virtual-Platform:~$

```

```

6598 Jps
deepakk@deepakk-VMware-Virtual-Platform:~$ start-dfs.sh
Jps
NC
56241 NodeManager
56135 ResourceManager
56703 Jps
deepakk@deepakk-VMware-Virtual-Platform:~$ ls -la /usr/local/hadoop/data/n
namenode
total 12
drwxr-xr-x 3 deepakk deepakk 4096 Aug 25 22:36 .
drwxr-xr-x 4 deepakk deepakk 4096 Aug 25 22:07 ..
drwx----- 2 deepakk deepakk 4096 Aug 25 22:36 current
deepakk@deepakk-VMware-Virtual-Platform:~$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [deepakk-VMware-Virtual-Platform]
deepakk@deepakk-VMware-Virtual-Platform:~$ jps
56241 NodeManager
57204 SecondaryNameNode
56918 NameNode
56135 ResourceManager
57388 Jps
57022 DataNode
deepakk@deepakk-VMware-Virtual-Platform:~$ start-yarn.sh
Starting resourcemanager
resourcemanager is running as process 56135. Stop it first and ensure /tmp
o/hadoop-deepakk-resourcemanager.pid file is empty before retry.
Starting nodemanagers
localhost: nodemanager is running as process 56241. Stop it first and ens
ure /tmp/hadoop-deepakk-nodemanager.pid file is empty before retry.
deepakk@deepakk-VMware-Virtual-Platform:~$ jps
57809 Jps
56241 NodeManager
57204 SecondaryNameNode
56918 NameNode
56135 ResourceManager
57022 DataNode
deepakk@deepakk-VMware-Virtual-Platform:~$ █

```



```
deepakk@deepakk-VMware-Virtual-Platform: ~  
56241 NodeManager  
57204 SecondaryNameNode  
56918 NameNode  
56135 ResourceManager  
57388 Jps  
57022 DataNode  
deepakk@deepakk-VMware-Virtual-Platform:~$ start-yarn.sh  
Starting resourcemanager  
resourcemanager is running as process 56135. Stop it first and ensure /tmp/hadoop-deepakk-resourcemanager.pid file is empty before retry.  
Starting nodemanagers  
localhost: nodemanager is running as process 56241. Stop it first and ensure /tmp/hadoop-deepakk-nodemanager.pid file is empty before retry.  
deepakk@deepakk-VMware-Virtual-Platform:~$ jps  
57809 Jps  
56241 NodeManager  
57204 SecondaryNameNode  
56918 NameNode  
56135 ResourceManager  
57022 DataNode  
deepakk@deepakk-VMware-Virtual-Platform:~$ hdfs dfs -mkdir /input  
deepakk@deepakk-VMware-Virtual-Platform:~$ hdfs dfs -ls  
ls: `.`: No such file or directory  
deepakk@deepakk-VMware-Virtual-Platform:~$ ^C  
deepakk@deepakk-VMware-Virtual-Platform:~$ hdfs dfs -ls /  
Found 1 items  
drwxr-xr-x - deepakk supergroup 0 2026-08-25 22:52 /input  
deepakk@deepakk-VMware-Virtual-Platform:~$ echo "Hadoop is a Big Data framework" > sample.txt  
deepakk@deepakk-VMware-Virtual-Platform:~$ echo "Hadoop is a Big Data framework" > sample.txt  
deepakk@deepakk-VMware-Virtual-Platform:~$ hdfs dfs -put sample.txt /input  
deepakk@deepakk-VMware-Virtual-Platform:~$ hdfs dfs -ls /input  
Found 1 items  
-rw-r--r-- 1 deepakk supergroup 31 2026-08-25 23:15 /input/sample.txt  
deepakk@deepakk-VMware-Virtual-Platform:~$ hdfs dfs -cat /input/sample.txt  
Hadoop is a Big Data framework  
deepakk@deepakk-VMware-Virtual-Platform:~$  
deepakk@deepakk-VMware-Virtual-Platform:~$ cd ~  
deepakk@deepakk-VMware-Virtual-Platform:~$ nano WordCount.java  
deepakk@deepakk-VMware-Virtual-Platform:~$
```

```
deepakk@deepakk-VMware-Virtual-Platform:~$ nano WordCount.java
GNU nano 8.7.1 WordCount.java
import java.io.IOException;
import java.util.StringTokenizer;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class WordCount {

    public static class TokenizerMapper
        extends Mapper<Object, Text, Text, IntWritable> {

        private final static IntWritable one = new IntWritable(1);
        private Text word = new Text();

        public void map(Object key, Text value, Context context)
            throws IOException, InterruptedException {

            StringTokenizer st = new StringTokenizer(value.toString());

            while (st.hasMoreTokens()) {
                word.set(st.nextToken());
                context.write(word, one);
            }
        }
    }

    public static class IntSumReducer
        extends Reducer<Text, IntWritable, Text, IntWritable> {

        [ Read 70 lines ]
        ^G Help      ^O Write Out    ^F Where Is     ^K Cut          ^T Execute
        ^X Exit      ^R Read File    ^\ Replace      ^U Paste        ^J Justify

deepakk@deepakk-VMware-Virtual-Platform:~$ hdfs dfs -put sample.txt /input
deepakk@deepakk-VMware-Virtual-Platform:~$ hdfs dfs -ls /input
Found 1 items
-rw-r--r--  1 deepakk supergroup          31 2026-08-25 23:15 /input/sample.txt
deepakk@deepakk-VMware-Virtual-Platform:~$ hdfs dfs -cat /input/sample.txt
Hadoop is a Big Data framework
deepakk@deepakk-VMware-Virtual-Platform:~$ cd ~
deepakk@deepakk-VMware-Virtual-Platform:~$ nano WordCount.java
deepakk@deepakk-VMware-Virtual-Platform:~$ nano WordCount.java
deepakk@deepakk-VMware-Virtual-Platform:~$ echo "hello hadoop hello bigdata
a hadoop" > input.txt
deepakk@deepakk-VMware-Virtual-Platform:~$ cat input.txt
hello hadoop hello bigdata hadoop
deepakk@deepakk-VMware-Virtual-Platform:~$ hdfs dfs -mkdir -p /wordcount/input
```

```
deepakk@deepakk-VMware-Virtual-Platform:~$ cat input.txt
hello hadoop hello bigdata hadoop
deepakk@deepakk-VMware-Virtual-Platform:~$ hdfs dfs -mkdir -p /wordcount/i
nput
deepakk@deepakk-VMware-Virtual-Platform:~$ hdfs dfs -put input.txt /wordco
unt/input
deepakk@deepakk-VMware-Virtual-Platform:~$ hdfs dfs -ls /wordcount/input
Found 1 items
-rw-r--r--  1 deepakk supergroup          34 2026-08-25 23:21 /wordcount/i
nput/input.txt
deepakk@deepakk-VMware-Virtual-Platform:~$
```

```
deepakk@deepakk-VMware-Virtual-Platform:~$ hdfs dfs -mkdir -p /wordcount/i
nput
deepakk@deepakk-VMware-Virtual-Platform:~$ hdfs dfs -put input.txt /wordco
unt/input
deepakk@deepakk-VMware-Virtual-Platform:~$ hdfs dfs -ls /wordcount/input
Found 1 items
-rw-r--r--  1 deepakk supergroup          34 2026-08-25 23:21 /wordcount/i
nput/input.txt
deepakk@deepakk-VMware-Virtual-Platform:~$ javac -classpath "$(hadoop clas
spath)" -d . WordCount.java
deepakk@deepakk-VMware-Virtual-Platform:~$ ls
Desktop      SnapshotTest      WordCount.java
Documents    Templates          hadoop-3.5.0.tar.gz
Downloads    Videos            hadoop-start.log
Music        'WordCount$IntSumReducer.class'  input.txt
Pictures     'WordCount$TokenizerMapper.class' sample.txt
Public       WordCount.class    snap
deepakk@deepakk-VMware-Virtual-Platform:~$ jar -cvf wordcount.jar WordCoun
t*.class
```

```
deepakk@deepakk-VMware-Virtual-Platform:~$ jar -cvf wordcount.jar WordCoun
t*.class
added manifest
adding: WordCount$IntSumReducer.class(in = 1657) (out= 708)(deflated 57%)
adding: WordCount$TokenizerMapper.class(in = 1752) (out= 769)(deflated 56%
)
adding: WordCount.class(in = 1472) (out= 814)(deflated 44%)
deepakk@deepakk-VMware-Virtual-Platform:~$
```



```
deepakk@deepakk-VMware-Virtual-Platform:~$ start-yarn.sh
Starting resourcemanager
Starting nodemanagers
deepakk@deepakk-VMware-Virtual-Platform:~$ hdfs dfs -ls /wordcount
Found 1 items
drwxr-xr-x   - deepakk supergroup          0 2026-08-25 23:21 /wordcount/input
deepakk@deepakk-VMware-Virtual-Platform:~$ hdfs dfs -rm -r /wordcount/output
rm: `/wordcount/output': No such file or directory
deepakk@deepakk-VMware-Virtual-Platform:~$ hadoop jar wordcount.jar WordCount /wordcount/input /wordcount/output
2026-08-25 23:35:32,870 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-08-25 23:35:39,647 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-08-25 23:35:40,211 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/deepakk/.staging/job_1787681065529_0001
2026-08-25 23:35:44,256 INFO input.FileInputFormat: Total input files to process : 1
2026-08-25 23:35:45,254 INFO mapreduce.JobSubmitter: number of splits:1
2026-08-25 23:35:47,415 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1787681065529_0001
2026-08-25 23:35:47,420 INFO mapreduce.JobSubmitter: Executing with tokens : []
2026-08-25 23:35:49,092 INFO conf.Configuration: resource-types.xml not found
2026-08-25 23:35:49,097 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-08-25 23:35:55,977 INFO impl.YarnClientImpl: Submitted application application_1787681065529_0001
deepakk@deepakk-VMware-Virtual-Platform:~$ hdfs dfs -cat /wordcount/output/part-r-00000
bigdata 1
hadoop 2
hello 2
deepakk@deepakk-VMware-Virtual-Platform:~$
```